7.1 Adding and Subtracting Polynomials

Name _____ Period ____ Date _____

Algebra I 7.1 (30-minute plan) · Student sheet

I can…

- find the degree of a monomial.
- write a polynomial in standard form and classify it by its degree and number of terms.
- add and subtract polynomials, including ones with two variables.

Vocabulary · fill in the missing word

Monomial — a number, a variable, or a product of a number and variables with _____ exponents

Degree of a monomial — the sum of the _____ of its variables

Polynomial — a monomial or a sum of _____

Binomial / trinomial — a _____ with two / three terms

Degree of a polynomial — the _____ degree of its terms

Standard form — terms in order of _____ exponents

Leading coefficient — the coefficient of the first term in _____ form

Part 1 · Notes and examples (fill in each note, then work the example)

Monomial: a number, a variable, or a _____ of a number and variables with _____ exponents.

Degree of a monomial: the _____ of the exponents of its variables.

Example 1

Find the degree of each monomial.

a. $\frac{1}{2}x^3$ b. $-3xy^4$ c. 8 d. $4x^2yz^3$

Polynomial: a monomial or a _____ of monomials. Binomial: ____ terms. Trinomial: ____ terms.

Degree of a polynomial: the _____ degree of its terms.

Standard form: exponents _____ from left to right. **Leading coefficient:** the coefficient of the _____ term.

Example 2

Write the polynomial in standard form. Identify the degree and leading coefficient. Then classify it by the number of terms.

$15x - x^3 + 3$

Subtracting polynomials: To subtract a polynomial, add its _____: change _____ sign.

Example 3

Find the difference.

$(4n^2 + 5) - (-2n^2 + 2n - 4)$

Part 2 · Partner practice

Partner practice

Find the sum.

$(3y^2 - 7y + 4) + (-5y^2 + y + 9)$